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Power control system

Abstract

A power control system for a forestry machine may include an engine control module and a pump control module. The engine control module may be configured to control an engine speed of an engine of the forestry machine. The engine speed may be limited in accordance with a maximum engine speed. The pump control module may determine whether a first function of the forestry machine is activated, determine whether a sensed pressure of a pump associated with the first function is greater than or equal to a high pressure setpoint, and output a control signal indicating a request for a power increase in response to determining that the first function is activated and the sensed pressure is greater than or equal to the high pressure setpoint. The engine control module may increase the maximum engine speed in response to the request for the power increase from the pump control module.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/169,133, filed on Mar. 31, 2021. The entire disclosure of the above application is incorporated herein by reference.

FIELD

(1) The present disclosure relates to power control systems for forestry machines.

BACKGROUND

(2) The background description provided here is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

(3) Forestry machines are configured to perform multiple functions related to topping, delimbing, cutting, and processing trees, removing tree stumps, etc. A forestry machine may include multiple attachments for performing respective functions. For example, a harvester may include multiple saws and attachments for performing delimbing, topping, and feed functions. A feller buncher is configured to grasp, cut, and move multiple trees at a time. A forestry excavator may be configured to perform harvester, processor, and loader functions, stump removal, etc.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIG. 1A is an example forestry machine according to the present disclosure;

(3) FIG. 1B is another example forestry machine according to the present disclosure;

(4) FIGS. 1C and 1D are example control sticks for controlling a forestry machine according to the present disclosure;

(5) FIG. 2 is a functional block diagram of a power control system according to the present disclosure; and

(6) FIG. 3 illustrates steps of an example power control method according to the present disclosure.

(7) In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

(8) Some functions performed by forestry machines require large amounts of power. Excavation functions, cutting functions, feed functions, etc. may encounter resistance during some tasks.

Accordingly, these tasks may require even greater amounts of power to complete. However, power is typically limited by the capabilities of a hydraulic system of the machine (e.g., pressure limits of the hydraulic system), which are in turn limited by engine speed and horsepower.

(9) Some machines may be configured to provide a temporary increase in engine speed and horsepower and a maximum pressure of the hydraulic system (i.e., a hydraulic pressure) to correspondingly increase power available to complete a task. For example, the machine may

include a button, switch, or lever that is activated to increase engine speed (and, correspondingly, a pump/hydraulic flow rate) and maximum pressure. In some examples, an operator of the machine must hold down a button to increase the engine speed and/or remove their hand from a control stick to activate a switch or lever. Accordingly, activating a power increase may complicate and interfere with efficient operation of the machine using the control sticks.

(10) Power control systems and methods for forestry machines according to the present disclosure are configured to automatically activate a power increase without requiring the operator to activate a separate button or switch. For example, the power increase may be activated in response to one or more selected functions (i.e., one of a subset of all possible functions performed by the machine, which may be referred to herein as “high pressure functions”) being activated and a pressure (e.g., a pressure of a pump associated with the task being performed) reaching a predetermined setpoint or threshold.

(11) Referring now to FIGS. 1A and 1B, an example forestry machine (e.g., a harvester **100**) configured to implement a power control system according to the present disclosure is shown. Although shown as a harvester, the principles of the present disclosure may also be applied to other forestry machines, such as a feller buncher, road building machines, etc. The harvester **100** includes a harvester head **104** configured to perform multiple forestry functions. For example, the harvester **100** may include multiple attachments including, but not limited to, front gripping knives or arms configured to grasp trees, a main saw, one or more secondary saws (e.g., a topping saw or blade), delimbing arms and blades, and feed rollers configured to grasp a tree trunk and roll (i.e., rotate) in forward and reverse directions to feed the tree through the harvester head **104**.

(12) The harvester head **104** is connected to a main body **108** of the harvester **100** via a boom and arm assembly **112**. A cab **116** is mounted on a frame above wheels **120** (as shown in FIG. 1A), tracks **122** (as shown in FIG. 1B), or a combination thereof. An operator controls functions of the harvester **100** from within the cab **116**. For example, the operator controls functions of the harvester head **104** and the boom and arm assembly **112** as well as an engine, hydraulic system, braking, etc. from within the cab **116**.

(13) Referring now to FIGS. 1B and 1C and with continued reference to FIG. 1A, the operator controls the harvester head **104** and the boom and arm assembly **112** using controls such as control sticks **124-1** and **124-2**, referred to collectively as control sticks **124**. For example only, the control sticks **124-1** and **124-2** correspond to left and right control sticks, respectively. Each of the control sticks **124** includes a plurality of actuators (e.g., switches and buttons), such as a rocker switch or trigger **128-1**, face buttons **128-2** (on respective faces **132** of the control sticks **124**), side console buttons **128-3** (on respective side consoles **136**), etc., referred to collectively as actuators **128**.

(14) The operator uses each of the actuators **128**, individually or collectively with another of the actuators **128**, to control respective attachments/functions of the harvester **100**. For example, the actuators **128** are configured to control functions including, but not limited to, a main saw, one or more secondary saws (e.g., a topping saw or blade), delimbing arms and blades, feed rollers (including separate actuators for controlling the feed rollers at different speeds and different feed directions), an orientation of the harvester head **104**, positioning of the boom and arm assembly **112**, gripping knives or arms, etc.

(15) Selected ones of the actuators **128** (i.e., a predetermined subset of all of the actuators **128**) are configured to activate a power increase (e.g., an engine speed and associated hydraulic pressure increase) without requiring the operator to activate a separate actuator as described below in more detail. The selected ones of the actuators **128** may correspond to predetermined forestry functions or attachments that may potentially require a power increase to complete associated tasks (i.e., high pressure functions).

(16) For example only, as described herein, the tasks that may trigger the power increase include, but are not limited to, a main saw of the harvester head **104**, feed forward and feed reverse functions (e.g., feed rollers) of the harvester head **104**, and a topping saw or blade of the harvester

head **104**. Conversely, actuators that control functions that do not typically require a power increase (e.g., delimbing arms and blades, movement of the harvester head **104** and/or boom and arm assembly **112**, movement of the forestry machine **100**, swing/rotation of an upper portion of the machine, etc.) are not configured to trigger the power increase. The power increase may be activated in response to a selected one of the actuators **128** being activated and a pressure (e.g., a hydraulic pressure of a pump associated with the task being performed) reaching a predetermined setpoint or threshold.

(17) Referring now to FIG. **2** and with continued reference to FIGS. **1A**, **1B**, and **1C**, an example power control system **200** according to the present disclosure is shown. The power control system **200** includes an optional attachment control module **204**, a pump control module **208**, an engine control module **212**, and a hydraulic system **220**. The engine control module **212** is configured to selectively increase a maximum engine speed of an engine **224** (e.g., a diesel or non-diesel internal combustion engine) to increase power provided to an attachment **228** of the forestry machine **100** in response to commands received from an operator at a user interface **232** as described below in more detail.

(18) The engine control module **212** is configured to control various functions associated with operation of the engine **224**, including combustion and injection timing, power demand and output (e.g., torque and horsepower output), engine speed, etc. Power output is a product of engine speed, which may be limited to a maximum engine speed to optimize fuel efficiency and minimize wear on the engine **224** and other components of the forestry machine **100**.

(19) As engine speed increases, a hydraulic flow (and, correspondingly, hydraulic power) that can be provided by the hydraulic system **220** also increases. For example, one or more pumps (e.g., a front pump, a rear pump, etc.) **236** of the hydraulic system **220** operate in accordance with the engine speed. Accordingly, if the engine control module **212** increases (e.g., conditionally and temporarily increases) the maximum engine speed, a maximum power available to the attachment **228** via the hydraulic system **220** also increases. For example, increasing the engine speed allows an increase in pump speed and/or pressure, hydraulic fluid flow rates, etc., and one or more valves (e.g., one or more valves **240** provided between an associated one of the pumps **236** and the attachment **228**, referred to hereinafter as “the valve” **240**) may be controlled to increase the pressure of the pump **236** and the corresponding pressure provided to the attachment **228**.

(20) For example, the operator of the forestry machine **100** controls the attachment **228** using the user interface **232** (e.g., one or more control sticks, such as the control sticks **124**). In this example, the attachment **228** corresponds to a function configured to trigger a power increase as described above, such as a main saw of the harvester head **104**, feed forward and feed reverse functions (e.g., feed rollers) of the harvester head **104**, a topping saw or blade of the harvester head **104**, etc. The user interface **232** outputs control signals (e.g., attachment control signals) to the attachment control module **204** in response to inputs from the operator, and the attachment control module **204** controls the attachment **228**. In some examples, the attachment control module **204** may be omitted and the user interface **232** outputs the control signals directly to the pump control module **208** and/or the engine control module **212**, which in turn control the attachment **228**.

(21) The pump control module **208** is further configured to communicate with the engine control module **212** and the hydraulic system **220**. For example, the pump control module **208** is configured to selectively implement the power increase according to the present disclosure. In this example, the pump control module **208** receives a signal from the attachment control module **204** indicating that the attachment **228** is being operated. In other words, the attachment control module **208** provides a signal to the pump control module **208** indicating that the operator is activating an actuator (e.g., a button) corresponding to the attachment **228**.

(22) In some examples, the attachment control module **204** generates the signal specifically in response to the activated function being a high pressure function. In other examples, the pump control module **208** receives a signal directly from the user interface **232** indicating that a high

pressure function was activated. In still other examples, the pump control module **208** receives a signal (directly from the user interface **232**, from the attachment control module **204**, etc.) when any function is activated and is configured to determine whether the activated function is a high pressure function. In any example, the signal received by the pump control module **208** is used to determine whether the function being activated is a high pressure function (i.e., one of a subset of all possible functions performed by the forestry machine **100**) that may trigger a power increase as described below in more detail.

(23) The pump control module **208** also receives signals from one or more pressure sensors **244**. For example, the pump control module **208** receives a signal from one of the pressure sensors **244** indicating a pressure (i.e., a sensed or measured pressure) of the pump **236** that provides flow to the attachment **228**. The pump control module **208** compares the sensed pressure of the pump **236** to a pressure setpoint or threshold (e.g., a high pressure setpoint). For example, the high pressure setpoint is determined in accordance with the operation of selected functions that may require a power increase. In other words, the selected functions that may require a power increase as described above may cause the pressure of the pump **236** to increase significantly. If the sensed pressure reaches the high pressure setpoint (e.g., due to a magnitude of a load being processed by the activated function) during the operation, the power increase may be required to complete the corresponding task.

(24) Accordingly, the pump control module **208** compares the sensed pressure to the high pressure setpoint to determine whether to trigger the power increase. If the pump control module **208** determines that sensed pressure has reached (i.e., is greater than or equal to) the high pressure setpoint and receives the signal indicating that a high pressure function of the attachment **228** is being operated, the pump control module **208** triggers the power increase.

(25) For example, to trigger the power increase, the pump control module **208** outputs a signal to the engine control module **212** indicating a request for the power increase and the engine control module **212** increases the maximum engine speed accordingly. In other words, during the power increase, the engine control module **212** allows the engine **224** to reach a higher maximum engine speed and power output (e.g., a maximum engine speed increase of 100-500 rpm and a power output increase of 10-50 horsepower). The pump control module **208** further increases a maximum pump pressure of one or more of the pumps **236** (e.g., by 100-500 psi).

(26) During the power increase, the valve **240** may be controlled to increase the pressure provided to the attachment **228**. For example, the pump control module **208** (or, in some examples, the attachment control module **204**) is configured to shift or adjust the valve **240**, to increase a maximum pressure that can be provided by the pump **236** in response the attachment **228** being activated and the sensed pressure reaching the high pressure setpoint. In other words, the pump control module **208** controls the valve **240** to achieve the increased pressure required by functions designated as high pressure functions. For example only, the valve **240** corresponds to a pressure relief valve.

(27) The pump control module **208** may end the power increase when one or more conditions are met. For example, the pump control module **208** may end the power increase (e.g., cause the engine control module **224** to decrease the maximum engine speed) in response to the operator releasing the actuator or button used to activate the attachment **228** and/or the sensed pressure decreasing below the high pressure setpoint.

(28) Further, a duration of the power increase may be limited regardless of whether the one or more conditions for ending the power increase are met. For example, the duration of the power increase may be limited to 2-15 seconds to minimize fuel consumption and component wear. In one example, the pump control module **208** initiates a timer **248** when the power increase is triggered. The timer **248** may be separate from or integral with (as shown) the pump control module **208**. When the timer reaches a predetermined time limit, the pump control module **208** ends the power increase. For example, the pump control module **208** outputs signals causing the engine control

module **224** to decrease the maximum engine speed

(29) Referring now to FIG. **3**, an example power control method **300** according to the present disclosure is shown. For example, the power control method **300** is implemented by the power control system **200** described above in FIG. **2**. At **304**, a forestry machine (e.g., the forestry machine **100**) is operated in a first mode. For example, in the first mode, engine speed is limited in accordance with a maximum engine speed and pump pressure is limited.

(30) At **308**, the method **300** (e.g., the attachment control module **204**) determines whether an operator of the forestry machine **100** has activated an actuator corresponding to a high pressure function of the attachment. If true, the method **300** continues to **312**. If false, the method **300** continues to **304**. At **312**, the method **300** (e.g., the pump control module **208**) determines whether a sensed pressure of a pump associated with the high pressure function or attachment has reached a high pressure setpoint. If true, the method **300** continues to **316**. If false, the method **300** continues to **304**.

(31) At **316**, the method **300** (e.g., the pump control module **208**) transitions the forestry machine **100** to a second mode and initiates a timer (e.g., the timer **248**). In the second mode, the maximum engine speed is increased and at least one valve is adjusted to allow a higher pump pressure to increase the power available to the attachment and the forestry machine **100**. At **320**, the forestry machine **100** is operated in the second mode.

(32) At **324**, the method **300** (e.g., the attachment control module **204** and/or the pump control module **208**) determines whether one or more conditions are met for ending the power increase and transitioning the forestry machine **100** to the first mode. For example, the method **300** determines whether the operator released the actuator or button used to activate the attachment **228** and/or the sensed pressure decreased below the high pressure setpoint. If true, the method **300** transitions the forestry machine **100** to the first mode and continues to **304**. If false, the method **300** continues to **328**.

(33) At **328**, the method **300** (e.g., the pump control module **208**) determines whether the timer **248** has reached a predetermined duration (i.e., whether a power increase duration has expired). If true, the method **300** transitions the forestry machine **100** to the first mode and continues to **304**. If false, the method **300** continues to **320**.

(34) The foregoing description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without altering the principles of the present disclosure. Further, although each of the embodiments is described above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of one or more embodiments with one another remain within the scope of this disclosure.

(35) Spatial and functional relationships between elements (for example, between modules, circuit elements, semiconductor layers, etc.) are described using various terms, including “connected,” “engaged,” “coupled,” “adjacent,” “next to,” “on top of,” “above,” “below,” and “disposed.” Unless explicitly described as being “direct,” when a relationship between first and second elements is described in the above disclosure, that relationship can be a direct relationship where no other intervening elements are present between the first and second elements, but can also be an indirect relationship where one or more intervening elements are present (either spatially or functionally) between the first and second elements. As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR,

and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

(36) In the figures, the direction of an arrow, as indicated by the arrowhead, generally demonstrates the flow of information (such as data or instructions) that is of interest to the illustration. For example, when element A and element B exchange a variety of information but information transmitted from element A to element B is relevant to the illustration, the arrow may point from element A to element B. This unidirectional arrow does not imply that no other information is transmitted from element B to element A. Further, for information sent from element A to element B, element B may send requests for, or receipt acknowledgements of, the information to element A.

(37) In this application, including the definitions below, the term “module” or the term “controller” may be replaced with the term “circuit.” The term “module” may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor circuit (shared, dedicated, or group) that executes code; a memory circuit (shared, dedicated, or group) that stores code executed by the processor circuit; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

(38) The module may include one or more interface circuits. In some examples, the interface circuits may include wired or wireless interfaces that are connected to a local area network (LAN), the Internet, a wide area network (WAN), a controller area network (CAN), or combinations thereof. The functionality of any given module of the present disclosure may be distributed among multiple modules that are connected via interface circuits. For example, multiple modules may allow load balancing. In a further example, a server (also known as remote, or cloud) module may accomplish some functionality on behalf of a client module.

(39) The term code, as used above, may include software, firmware, and/or microcode, and may refer to programs, routines, functions, classes, data structures, and/or objects. The term shared processor circuit encompasses a single processor circuit that executes some or all code from multiple modules. The term group processor circuit encompasses a processor circuit that, in combination with additional processor circuits, executes some or all code from one or more modules. References to multiple processor circuits encompass multiple processor circuits on discrete dies, multiple processor circuits on a single die, multiple cores of a single processor circuit, multiple threads of a single processor circuit, or a combination of the above. The term shared memory circuit encompasses a single memory circuit that stores some or all code from multiple modules. The term group memory circuit encompasses a memory circuit that, in combination with additional memories, stores some or all code from one or more modules.

(40) The term memory circuit is a subset of the term computer-readable medium. The term computer-readable medium, as used herein, does not encompass transitory electrical or electromagnetic signals propagating through a medium (such as on a carrier wave); the term computer-readable medium may therefore be considered tangible and non-transitory. Non-limiting examples of a non-transitory, tangible computer-readable medium are nonvolatile memory circuits (such as a flash memory circuit, an erasable programmable read-only memory circuit, or a mask read-only memory circuit), volatile memory circuits (such as a static random access memory circuit or a dynamic random access memory circuit), magnetic storage media (such as an analog or digital magnetic tape or a hard disk drive), and optical storage media (such as a CD, a DVD, or a Blu-ray Disc).

(41) The apparatuses and methods described in this application may be partially or fully implemented by a special purpose computer created by configuring a general purpose computer to execute one or more particular functions embodied in computer programs. The functional blocks, flowchart components, and other elements described above serve as software specifications, which can be translated into the computer programs by the routine work of a skilled technician or programmer.

(42) The computer programs include processor-executable instructions that are stored on at least one non-transitory, tangible computer-readable medium. The computer programs may also include or rely on stored data. The computer programs may encompass a basic input/output system (BIOS) that interacts with hardware of the special purpose computer, device drivers that interact with particular devices of the special purpose computer, one or more operating systems, user applications, background services, background applications, etc.

(43) The computer programs may include: (i) descriptive text to be parsed, such as HTML (hypertext markup language), XML (extensible markup language), or JSON (JavaScript Object Notation) (ii) assembly code, (iii) object code generated from source code by a compiler, (iv) source code for execution by an interpreter, (v) source code for compilation and execution by a just-in-time compiler, etc. As examples only, source code may be written using syntax from languages including C, C++, C#, Objective-C, Swift, Haskell, Go, SQL, R, Lisp, Java®, Fortran, Perl, Pascal, Curl, OCaml, Javascript®, HTML5 (Hypertext Markup Language 5th revision), Ada, ASP (Active Server Pages), PHP (PHP: Hypertext Preprocessor), Scala, Eiffel, Smalltalk, Erlang, Ruby, Flash®, Visual Basic®, Lua, MATLAB, SIMULINK, and Python®.

Claims

1. A power control system for a forestry machine configured to perform a plurality of functions, the power control system comprising: an engine control module configured to control an engine speed of an engine of the forestry machine, wherein the engine speed is limited in accordance with a maximum engine speed; and a pump control module configured to: determine whether a first function of a predetermined subset of the plurality of functions of the forestry machine is activated, wherein the predetermined subset of the plurality of functions are functions of one or more selected attachments of the forestry machine; determine whether the activated first function is designated as a high pressure function; determine that a magnitude of a pressure of a pump that provides flow to the activated first function has increased, wherein the increase of magnitude of the pressure is caused by the activation of the first function; determine that the magnitude of the pressure is greater than or equal to a high pressure setpoint; and upon determining that the first function is activated, the activated first function is a high pressure function, the magnitude of the pressure of the pump that provides flow to the activated first function has increased, and the increase in the pressure is greater than or equal to the high pressure setpoint, output a control signal indicating a request for a power increase, wherein the engine control module receives the request for the power increase and is configured to increase the maximum engine speed of the engine in response to the request for the power increase received from the pump control module.
2. The power control system of claim 1, wherein determining whether the first function is activated includes receiving a signal indicating that a first attachment configured to perform the first function is activated.
3. The power control system of claim 1, wherein the pump control module is configured to initiate a timer upon detecting the power increase and end the power increase when the timer reaches a predetermined duration.
4. The power control system of claim 1, wherein the pump control module is configured to end the power increase in response to a determination that at least one of (i) the first function is no longer activated and (ii) the increase in the pressure is less than the high pressure setpoint.
5. The power control system of claim 1, further comprising a valve that is selectively shifted or adjusted to increase pressure output by the pump in response to the first function being activated and the increase in the pressure being greater than or equal to the high pressure setpoint.
6. The power control system of claim 5, further comprising an attachment control module.
7. The power control system of claim 6, wherein the attachment control module is configured to receive attachment control signals from a user interface.

8. The power control system of claim 7, wherein the user interface includes at least one control stick of the forestry machine.
 9. The power control system of claim 8, wherein determining whether the first function is activated includes determining whether an actuator on the at least one control stick that is used to control the first function is activated.
 10. A forestry machine comprising the power control system of claim 1.
 11. The forestry machine of claim 9, wherein the forestry machine is one of a harvester, a forestry excavator, and a feller buncher.
 12. A method of operating a power control system for a forestry machine configured to perform a plurality of functions, method comprising: controlling an engine speed of an engine of the forestry machine, wherein the engine speed is limited in accordance with a maximum engine speed; determining whether a first function of a predetermined subset among the plurality of functions of the forestry machine is activated, wherein the predetermined subset of the plurality of functions are functions of one or more selected attachments of the forestry machine; determining whether the activated first function is designated as a high pressure function; determining that a magnitude of a pressure of a pump that provides flow to the activated first function has increased, wherein the increase of magnitude of the pressure is caused by the activation of the first function; determining that the magnitude of the pressure is greater than or equal to a high pressure setpoint; upon determining that the first function is activated, the activated first function is a high pressure function, the magnitude of the pressure of the pump that provides flow to the activated first function has increased, and the increase in the pressure is greater than or equal to the high pressure setpoint, output a control signal indicating a request for a power increase; and increasing the maximum engine speed of the engine in response to the request for the power increase.
 13. The method of claim 12, wherein determining whether the activated first function includes receiving a signal indicating that a first attachment configured to perform the activated first function.
 14. The method of claim 12, further comprising initiating a timer when the power increase is detected and ending the power increase when the timer reaches a predetermined duration.
 15. The method of claim 12, further comprising ending the power increase in response to a determination that at least one of (i) the first function is no longer activated and (ii) the increase in the pressure is less than the high pressure setpoint.
 16. The method of claim 12, further comprising selectively shifting a valve to increase pressure output by the pump in response to the first function being activated and the increase in the pressure being greater than or equal to the high pressure setpoint.
 17. The method of claim 12, further comprising receiving attachment control signals from a user interface, wherein the user interface includes at least one control stick of the forestry machine.
 18. The method of claim 17, wherein determining whether the activated first function includes determining whether an actuator on the at least one control stick that is used to control the activated first function.
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